

Atari 400/800 — Chip Pinouts

MAJOR SOCKETED ICS · 6502 · ANTIC · GTIA · POKEY · PIA · RAM · TEST & REPAIR

DIP pinouts seen from the top, notch up, pin 1 at top-left. See the **Legend** card for notation & colour key. **F00**=fast φ0 out.

Legend — reading the pinouts notation

/NAMEActive LOW — true / asserted when the pin is at logic **LOW** (~0 V). A leading / (datasheets often use an overline) marks these: e.g. /RES /IRQ /CS /WE.

NAME Active HIGH / normal — asserted when the pin is **HIGH** (~+5 V). No slash. R/W is a level: HIGH = read, LOW = write.

Colour key: **Vcc / Vdd = power** · **GND / Vss = ground** · **φ = clock**. **Orientation:** chip seen from the top, notch up — pin 1 top-left, numbers run **down the left** then **up the right**.

6502 CPU 40-DIP

Vss	1	40	/RES
RDY	2	39	φ2 out
φ1 out	3	38	S0
/IRQ	4	37	φ0 in
NC	5	36	NC
/NMI	6	35	NC
SYNC	7	34	R/W
Vcc	8	33	D0
A0	9	32	D1
A1	10	31	D2
A2	11	30	D3
A3	12	29	D4
A4	13	28	D5
A5	14	27	D6
A6	15	26	D7
A7	16	25	A15
A8	17	24	A14
A9	18	23	A13
A10	19	22	A12
A11	20	21	Vss

Atari CPU @ 1.79 MHz. ANTIC can **HALT** it to steal cycles for the display. **Fails:** dead machine, black or red screen.

ANTIC display proc 40-DIP

Vss	1	40	D4
AN0	2	39	D5
AN1	3	38	D6
/LP	4	37	D7
AN2	5	36	/RST
/RNMI	6	35	F00
/NMI	7	34	φ0
/REF	8	33	D3
/HALT	9	32	D2
A3	10	31	D1
A2	11	30	D0
A1	12	29	φ2
A0	13	28	A4
R/W	14	27	A5
RDY	15	26	A6
A10	16	25	A7
A12	17	24	A8
A13	18	23	A9
A14	19	22	A11
A15	20	21	Vcc

Display-list processor: fetches the screen, drives **HALT**/RDY for DMA, generates /NMI (VBlank/DLI) and feeds AN0-AN2 to GTIA. **Fails:** no sync / blank or scrambled display.

GTIA colour/sprites 40-DIP

A1	1	40	A2
A0	2	39	A3
Vcc	3	38	A4
D3	4	37	D4
D2	5	36	D5
D1	6	35	D6
D0	7	34	D7
T0	8	33	R/W
T1	9	32	/CS
T2	10	31	LUM0
T3	11	30	φ2
S0	12	29	F00
S1	13	28	OSC
S2	14	27	Vss
S3	15	26	HALT
PAL	16	25	CSYNC
CAD3	17	24	LUM3
AN0	18	23	LUM2
AN1	19	22	LUM1
AN2	20	21	COLOR

Colour + player/missile (sprite) generator. Turns ANTIC's AN0-AN2 into LUM/COLOR + CSYNC, reads triggers T0-T3. C014805=NTSC, C014889=PAL, early C012295=CTIA. **Fails:** wrong/no colour, missing sprites.

POKEY sound/I-O 40-DIP

Vss	1	40	D2
D3	2	39	D1
D4	3	38	D0
D5	4	37	AUDIO
D6	5	36	A0
D7	6	35	A1
φ2	7	34	A2
P6	8	33	A3
P7	9	32	R/W
P4	10	31	CS1
P5	11	30	/CS0
P2	12	29	/IRQ
P3	13	28	S0D
P0	14	27	OCLK
P1	15	26	BCLK
/KR2	16	25	/KR1
Vdd	17	24	SID
/K5	18	23	/K0
/K4	19	22	/K1
/K3	20	21	/K2

Sound (4 voices) + keyboard scan (K0-K5) + paddle pots + SIO serial. **Fails:** one dead voice, no keyboard, or SIO won't work at fast divisors.

6520 PIA 40-DIP

Vss	1	40	CA1
PA0	2	39	CA2
PA1	3	38	/IRQA
PA2	4	37	/IRQB
PA3	5	36	RS0
PA4	6	35	RS1
PA5	7	34	/RES
PA6	8	33	D0
PA7	9	32	D1
PB0	10	31	D2
PB1	11	30	D3
PB2	12	29	D4
PB3	13	28	D5
PB4	14	27	D6
PB5	15	26	D7
PB6	16	25	φ2/E
PB7	17	24	CS1
CB1	18	23	/CS2
CB2	19	22	CS0
Vcc	20	21	R/W

\$D300: joystick ports, SIO control lines (CMD, MOTOR, PROCEED), bank/OS control. Any 6520 / 6820 / 6821 works. **Fails:** dead joysticks, no disk/SIO.

Memory & ROM RAM/ROM

RAM	4116 DRAM (16K×1, ±12/5/-5V)
OS ROM	10/16KB at \$C000/\$D800
BASIC	8KB cartridge / built-in (XL)
MMU/glue	decode + 74-series

Fails: red screen + memo-test squares = bad 4116. A bad OS ROM stops boot; BASIC ROM can be pulled.

Test & Repair tips

Red screen	reseat / swap 4116 RAM
No display	ANTIC ↔ GTIA path
No colour	GTIA
Dead keys	POKEY
Bad sound/SIO	POKEY
No joystick/disk	6520 PIA

Atari used cheap single-wipe RAM sockets — reseal all chips first. ANTIC talks to GTIA through the 4050 buffer.